EQUIPMENT AND SITE SELECTION

Introduction

Choosing the proper equipment and site for blood collection differs depending on the type of test being performed and the age and condition of the patient. Reviewing the requisition instructs you on the method to acquire the test results and how to share them with the patient. Different tests and patient conditions may require different supplies, positioning, and procedures. The supplies chosen must also be observed to confirm that their current condition meets the standards required for the test being performed. Choosing the appropriate collection site helps reduce the risk of injury to the patient and increase the likelihood of accurate test results.

Equipment and Supplies



Attention to choosing the proper equipment helps reduce the risk of injury while increasing the likelihood of test results that reflect the patient's condition.

Review the requisition to assure that you are gathering the proper supplies and familiarize yourself with any patient requirements such as fasting or allergies to latex.

Confirm that your phlebotomy tray or cart has the supplies necessary to perform the test ordered. It can delay a blood collection or even increase the risk of cross-contamination if you do not have all of the supplies when you enter the patient's area. It also appears unprofessional to have to leave the room because you forgot an essential supply.

Routine venipunctures are performed by the evacuated tube system. This method of blood collection consists of a double-ended needle, vacuum tubes, and a tube holder that helps hold the needle in place, allowing tubes to be slid in and attached to the needle.

When performing a venipuncture with the evacuated tube system, gather the following supplies.

- Tourniquet: Disposable, blood pressure cuff, or multiple use
- Needle: Double-ended safety needle
- Tube holder: Disposable single use
- Gauze: Two pieces

- Bandage: Tape, adhesive bandage, or Coban (an adhesive wrap like an elastic bandage that adheres to itself)
- Alcohol swaps: Individually wrapped alcohol pads
- Sharps container: Confirm that it is not filled above the fill line
- Vacuum blood collection tubes: Color of tube(s) chosen to correspond with the test ordered

Along with the necessary supplies for venipuncture, you also need the required personal protective equipment (PPE). Some of the requirements for PPE may be listed on the requisition. Follow all of your facility's requirements regarding PPE.

Tourniquets

The tourniquet can be either single use or multiple use depending on the requirements of your facility and the type needed for the collection. Most tourniquets are made of stretchable material and are single use and disposable. Before use, they should be inspected for any tears or other flaws. Multiple-use tourniquets such as ones with a hook-and-loop fastener should be inspected for tears, blood, or other stains and dirt. Do not use it if it is dirty or torn. They must be cleaned between each patient. Sometimes, a blood pressure cuff is used instead of a tourniquet. When a blood pressure cuff is used, confirm that the hook-and-loop fastener is intact and the bulb pumps easily and releases smoothly. When using a blood pressure cuff for a tourniquet, inflate to 40 mm Hg and release at the same time a tourniquet would be released.

To help find the vein and provide accurate test results, a tourniquet is applied 3 to 4 inches above the venipuncture site. Once the tourniquet is tied on, it should be removed in less than 1 min. When the tourniquet is in place for longer than that, it causes ***hemoconcentration***. This excessive collection of blood in the area can not only cause injury to your patient by pressing on nerves surrounding the needle entry site but also cause the blood test to demonstrate a higher level of the components that are being tested.

hemoconcentration. An excess of blood accumulated in an area of body.

TABLE 1.1 Types of Tourniquets

TYPE	USAGE	MAINTENANCE
Disposable single-use	Tie snugly 3 to 4 inches above the site.	Dispose after every patient. Confirm that there are no rips or tears.
Blood pressure cuff	Place snugly 3 inches above the site and pump to 40 mm Hg.	Confirm that the bulb is functional. Clean after every use.
Multiple-use/hook-and-loop fastener or buckle	Attach hook to loop or buckle snugly 3 to 4 inches above the site.	Confirm that the hook-and-loop fastener is operable, there are no tears, and the surface is free from dirt and debris. Clean after each use.

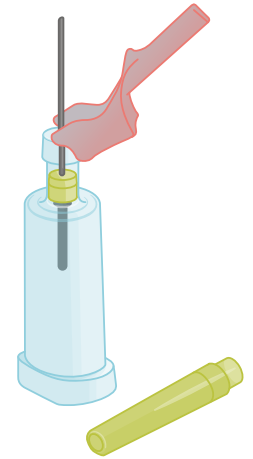
Venipuncture Needles

Another necessary venipuncture supply is the needle. There are two routine types of venipuncture needles. Straight needles are also called double-ended needles, safety needles, or eclipse needles and are used for routine blood collection. Another type is a winged infusion set, also called a butterfly, used for small, difficult-to-access veins. It has tubing that connects the needle inserted into the patient to the needle attached to the adapter, which is secured when the adapter is connected to the needle assembly.

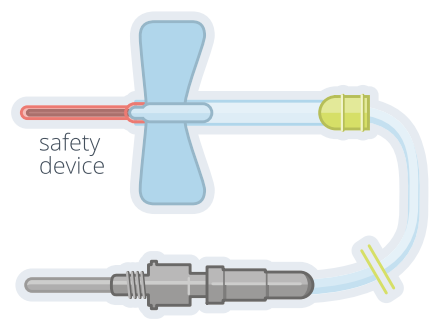
Both needles are double ended, with one end, which has beveled point at the end, inserted into the patient. The other end consists of a sharp needle covered by a rubber sheath. This sheath pushes back when the needle slides through the tube stopper. The vacuum in the tube pulls the blood through the needle.

In some situations, a patient may have veins that are very fragile or collapse easily. The force of the vacuum in an evacuated tube may collapse these veins, making it impossible to collect a sample. For these situations, a syringe system may be used. A syringe needle is similar in appearance to the evacuated tube system, a beveled straight needle that has a safety. The gauge is similar, as well, ranging from 21 to 23 gauge. Syringe needles are single-ended needles, not double like butterfly and routine venipuncture needles. Instead, they have one needle with a Luer-Lok base that screws into the adapter.

Double-ended straight needle



Butterfly needle



Syringe

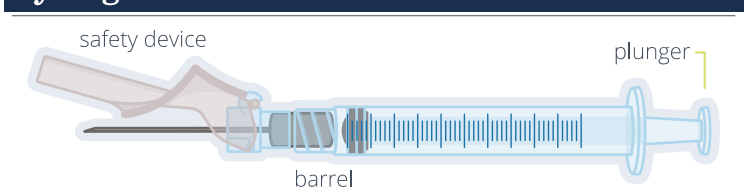


TABLE 1.2 Types of Venipuncture Needles

NEEDLE TYPE	TYPICAL SIZES	LENGTHS	TYPICAL USES
Double-ended straight needle	21- to 23-gauge	1 to 1 1/2 inches	Routine venipuncture
Butterfly needle	21- to 23-gauge	1/2 to 3/4 inch	Small, difficult-to-access veins
Syringe	21- to 23-gauge	1 to 1 1/2 inches	Fragile, easily-collapsible veins

Venipuncture needles are measured in gauges, typically from 21 to 25 gauge, with 21 gauge used for most routine venipunctures. If a patient has smaller veins, a 22-gauge needle is frequently used. Other blood collection needles, like those used for blood donation, are larger, typically 16 to 18 gauge. The smaller the gauge, the larger the diameter of the needle. For example, a 16-gauge needle has a larger diameter than a 21-gauge needle. In certain circumstances, a 25-gauge needle may be used, such as in infants and children, difficult-to-access adult veins, or veins in the hand. Using such a small needle can ***hemolyze*** the sample, resulting in inaccurate test results. A hemolyzed sample may be rejected by the laboratory, and blood may need to be recollected. It can also increase the time it takes to withdraw the sample, especially when using a butterfly. Clots can also form in the tubing, causing the blood flow to stop.

While the tube choice is made when gathering your supplies, your needle is chosen when you determine your patient's vein condition. Bring all sizes of needles, both butterfly and straight, to collect blood from your patient. You will determine which one you will use based on the patient's vein size and location. There will be more on this in the next lesson.

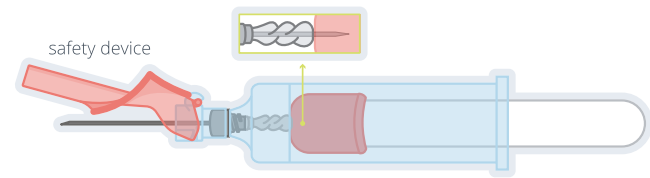
Use the Best Needle Size, Every Time

A needle that is too large unnecessarily hurts the patient, while a needle that is too small can destroy blood cells.

hemolyze. The destruction of red blood cells; can negatively impact samples.

Tube Holders

Tube holders are single-use items that attach to the needle to assist in blood collection. Tube holders have one side where the needle is attached and another, larger side where the vacuum tube is inserted. Unlike other venipuncture supplies, the color has no particular significance to the collection. Before using the tube holder, confirm that it is not cracked or dirty.



Alcohol Pads and Gauze

Alcohol pads are single use and individually packaged. The packaging should be observed for their expiration date, and the user should verify the wrapper is not ripped or torn. Typically, 70% isopropyl or ethyl alcohol pads are used. For blood cultures, 1% to 10% povidone-iodine swab sticks or chlorhexidine gluconate are used. Nonalcohol-based cleanser must be used for blood alcohol specimens. Choose two gauze pads. Gauze does not have to be individually wrapped, but it is important to confirm that it is clean and does not have any visual signs of dirt or other debris. Cotton balls should not be used as they can rip off a scab or leave fibers on the insertion site.

Sharps Containers



Sharps containers must be leak resistant and puncture proof and have a locking lid. They should also have a biohazard label displayed in an easily observed location. Only sharp items such as needles should be disposed of into these containers. Do not use a sharps container that is filled past the fill line.

Bandages

Confirm that you have several types of bandages in your cart or on your tray. Routine venipunctures may use regular adhesive bandages, similar to what is put on the site after an injection. A patient with delicate or fragile skin may be better bandaged with Coban. Another bandaging option is gauze and tape.

Vacuum Tubes

Vacuum tubes are used to collect blood for testing. Each tube has a measured amount of vacuum, calculated to collect a determined amount of blood. This measurement is printed on the side of the tube in milliliters. The tubes can also contain additives that have various functions such as preventing clots, preserving blood components, or activating clotting. Each additive impacts the blood in a different manner, so it is critical to use the proper tube for the test ordered. Reviewing the requisition helps you determine which tubes should be selected for the test being performed.

Blood collection tubes contain a vacuum that helps draw the proper amount of blood. Each tube that contains an additive has a unique stopper color that identifies the additive placed in the tube. These additives can be in the form of liquid or powder and also have an impact on what type of sample they create when collected from the body and centrifuged. Typically red, gold, or serum separator tubes will yield serum, while most other tubes yield plasma. Serum tubes need to clot for 30 to 60 minutes before being centrifuged. Plasma may be centrifuged immediately after collection. After being centrifuged, both serum and plasma samples will look the same, so it is critical to use the proper tube for every venipuncture.

Tubes are collected in a particular order called the order of draw. Attached to the double-ended needle, the side that enters the tube has a cover that slides up when the tube is attached. When the next tube is attached, some of the additives can be carried from the tube previously collected to the next tube. The laboratory has their equipment calibrated to process tests in a particular order. When a tube is collected in the improper order, the additional additives may change not only how the blood is preserved but also the blood test results. This can negatively impact your patient's diagnosis and treatment. It is important to always follow the order of draw specified by your facility's laboratory.

TABLE 1.3 The Order of Draw of Commonly Used Tubes

VACUUM TUBE COLOR	ADDITIVE	LABORATORY USE	NUMBER OF INVERSIONS
Yellow-top tube or blood culture bottles	• Sodium polyanethol sulfonate • Prevents blood from clotting and stabilizes bacterial growth	Blood or body fluid cultures	-
Light blue	• Sodium citrate • Removes calcium to prevent blood from clotting	Coagulation testing	3 to 4
Red	None	• Serum test • Chemistry studies • Blood bank • Immunology	5

TABLE 1.3 The Order of Draw of Commonly Used Tubes

VACUUM TUBE COLOR	ADDITIVE	LABORATORY USE	NUMBER OF INVERSIONS
Red/gray marbled	• No anticoagulant but contains silica particles to enhance clot formation • For serum separation	• Serum test • Chemistry studies • Immunology	5
Green	• Heparin • Inhibits thrombin formation to prevent clotting	Chemistry test	8
Green/gray marble	• Lithium heparin and gel • For plasma separation	Plasma determinations in chemistry studies	8
Lavender	• Ethylenediaminetetraacetic acid (EDTA) • Removes calcium to prevent blood from clotting	Hematology test	8
Gray	• Potassium oxalate and sodium fluoride • Removes calcium to prevent blood from clotting • Fluoride inhibits glycolysis	Chemistry testing, especially glucose and alcohol levels	8 to 10

**Always follow manufacturer recommendations when determining the order of draw.*

When choosing the appropriate tubes for your blood collection, also check them for their expiration date. The vacuum is reduced in expired tubes, which could result in a lower volume of blood collected. Another issue with expired tubes is that the additive may be expired. Expired additives may adversely affect the test results. Both issues can contribute to a patient having to endure more testing or receiving an inaccurate diagnosis and treatment because of test results that are not in range.

Do Not Use Expired Vacuum Tubes

When a tube expires, the amount of vacuum and quality of the additives are reduced.

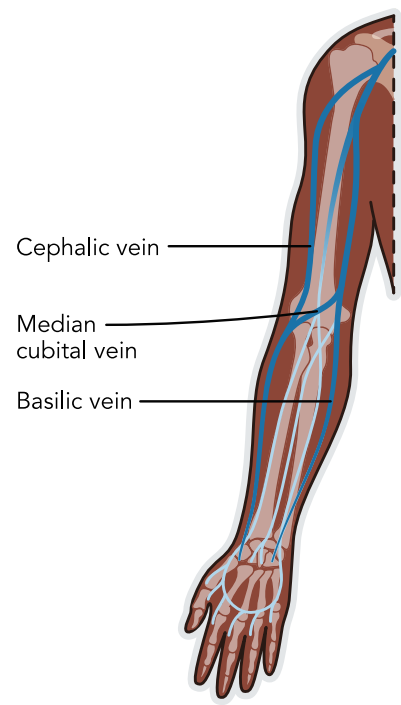
Besides the expiration date, tubes should be examined to confirm that there are not cracks or splits in the tube or stopper. It is also necessary to confirm that the tubes will hold enough blood for the tests to be performed. Each blood test requires a certain amount of blood to help assure the accuracy of the test. For example, if the tests require 11 mL of blood and the tube you need comes only in 6-mL sizes, you will use two tubes for that collection.

Site Selection in the Antecubital Region

Blood collected by venipuncture uses the veins in the ***antecubital*** region of the arm and the dorsal part of the hand. The antecubital region is located slightly below the bend of the elbow, anterior to the elbow joint.

Knowing and being able to find the best vein for venipuncture is one of the most important steps of performing this type of blood collection. Venipuncture can be performed in the antecubital region of the arm and the dorsal part of the hand. In rare cases and with permission from the provider, the foot or ankle may be used.

The first choice for a routine venipuncture is the antecubital region of the arm. To help place your patient in position, once they are seated or reclined, ask them to extend their arm out straight, palm up. This will reveal the antecubital region, located slightly below the bend of the elbow. This is the area on the other side of the elbow about midway from the hand to the shoulder. If you are unsure where this region is on your patient, place your thumb on their elbow and wrap your hand around the arm. Where your fingers land is the antecubital area on your patient.



antecubital. The inside of the elbow; location of the veins used in venipuncture.

median cubital. Vein near the middle of antecubital region; first choice for venipuncture.

cephalic. Vein located in the antecubital, near the outside of the arm; second choice for venipuncture.

basilic. Vein located in the antecubital near the body; the last choice for venipuncture.

The veins typically used for venipuncture in this area include the ***median cubital***, ***cephalic***, and ***basilic***. The median cubital vein is the best choice for venipuncture as it is located near the antecubital region. Its location also helps it stay better anchored in place in the arm. The cephalic vein is located closer to the outer side of the arm and is the second-best choice. The basilic vein is located closer to the inner side of the body. Because this vein is located close to the median nerve and brachial artery and rolls more easily, a needle miss in this area could harm the patient. This vein should only be used if you cannot find a vein suitable for venipuncture in either arm or hand.